

POSTS AND TELECOMMUNICATIONS INSTITUTE OF
TECHNOLOGY
Faculty of Information Technology

Intelligent System Development
ASSIGNMENT 02

From Data Representation to a Deployable Intelligent System

"A model is only as reliable as the data and evaluation process behind it."

Instructor / Lecturer: Assoc. Prof. Dinh Que Tran, Ph.D.
Email: quetd@ptit.edu.vn, tdque@yahoo.com

Student Name: Lưu Anh Dũng
Student ID: B23DCDK036
Semester: I.2025 – 2026
Class: E23CNPM02
Submission: Individual Project
Repository: github.com/dzungluu25/Intelligence_System

Contents

1	Executive Summary	2
2	Connection to Lecture 02: Data Representation	3
3	Application 1 — Diabetes Prediction	4
3.1	Problem Description	4
3.2	Kaggle Dataset	4
3.3	Data Understanding	4
3.4	Data Cleaning	4
3.5	Data Representation	4
3.6	Exploratory Data Analysis	4
3.7	Model Development	5
3.8	Evaluation	5
3.9	Model Selection	5
3.10	Deployment	5
4	Application 2 — House Price Prediction	5
4.1	Problem Description	5
4.2	Dataset	6
4.3	Data Understanding and Cleaning	6
4.4	Representation	6
4.5	Exploratory Data Analysis	6
4.6	Regression Models and Evaluation	6
4.7	Model Selection and Deployment	6
5	Application 3 — E-Commerce Customer Behavior	6
5.1	Problem Description	6
5.2	Dataset	7
5.3	Customer Representation: Multimodal Pipeline	7
5.4	Data Leakage Prevention	7
5.5	Model Development and Evaluation	7
5.6	Business Interpretation and Deployment	7
6	Comparison of the Three Intelligent Systems	8
6.1	Cross-Application Benchmark	8
6.2	Comparative Analysis Questions	8
7	Deployment Architecture	8
8	Web Application	9
9	Mobile Application	9
10	Deployment and Inference	9
11	Reproducibility	9
12	Final Comparative Discussion	10
13	Conclusion	10
A	Repository Structure	11
B	Required Notebook Structure	11

C	Required API Structure	12
D	Web Application Interface	12
E	Mobile Application Interface	14

1 Executive Summary

This report builds three deployable intelligent systems from raw Kaggle data: a diabetes screening classifier, a house-price regressor, and an e-commerce product-recommendation classifier that incorporates review text. Each system is developed through an end-to-end pipeline and exposed as a FastAPI POST `/predict` microservice, consumed by both a modern React/Vite web interface and a Flutter mobile application.

Application	Prediction / Task	Main Representation
Diabetes	Diabetes classification	Scaled & passthrough feature matrix
House Price	Price regression	Imputed, scaled & one-hot matrix
Customer Behavior	Behavior / interest prediction	Tabular matrix + sparse TF-IDF vectors

Table 1: Overview of the Three Intelligent Systems

1. Diabetes Prediction

- **Dataset:** Diabetes Health Indicators (CDC BRFSS 2015).
- **Problem:** Binary classification – predict whether a survey respondent has diabetes.
- **Representation:** Scaled continuous features + passthrough binary/ordinal features: $X \in \mathbb{R}^{N \times 23}$.
- **Selected Model:** Random Forest Classifier.
- **Main Evaluation Result:** ROC-AUC: 0.809, Recall(1): 0.735, Precision(1): 0.357.
- **Deployment Method:** FastAPI backend + React/Vite web client + Flutter mobile app.

2. House Price Prediction

- **Dataset:** Vietnamese Real Estate Listings (Apr–Sept 2025).
- **Problem:** Regression – predict the property sale price ($\log(1 + y)$ in million VND).
- **Representation:** Imputed + scaled continuous features + one-hot encoded categorical features: $X \in \mathbb{R}^{N \times 92}$.
- **Selected Model:** Random Forest Regressor.
- **Main Evaluation Result:** RMSE: 26,873 (million VND), MAE: 10,626, R^2 : 0.186.
- **Deployment Method:** FastAPI backend + React/Vite web client + Flutter mobile app.

3. E-Commerce Customer Behavior

- **Dataset:** Sephora Products and Skincare Reviews.
- **Problem:** Binary classification – predict whether a customer will recommend the product.
- **Representation:** Tabular one-hot and scaled matrix ($X_{\text{tab}} \in \mathbb{R}^{N \times 138}$) + sparse TF-IDF text features ($X_{\text{text}} \in \mathbb{R}^{N \times 28,595}$).
- **Selected Model:** Logistic Regression on combined tabular + TF-IDF text representations.
- **Main Evaluation Result:** ROC-AUC: 0.964, Macro-F1: 0.858, Recall(0): 0.876, Accuracy: 0.918.
- **Deployment Method:** FastAPI backend + React/Vite web client + Flutter mobile app.

2 Connection to Lecture 02: Data Representation

A central premise of Lecture 02 is that machine learning models cannot process raw real-world entities directly; they require structured mathematical representations:

Real-world object $\longrightarrow$ Raw data $\longrightarrow$ Numerical representation $\longrightarrow$ Tensor / Matrix

Application	Raw Data Form	ML Computational Representation
Diabetes	CSV (survey responses)	Dense feature matrix $X \in \mathbb{R}^{N \times 23}$
House Price	CSV (scraped listings)	Imputed/encoded matrix $X \in \mathbb{R}^{N \times 92}$
Customer Behavior	CSV (user profile + review)	Tabular matrix + sparse TF-IDF text vectors

Table 2: Transformation from Raw Data to Mathematical Representation

1. What does one row represent?

- *Diabetes*: One individual BRFSS telephone survey respondent.
- *House Price*: One specific real estate property listing.
- *Customer Behavior*: One user review transaction for a Sephora skincare item.

2. What does one column represent? A single observable attribute, engineered indicator, token/n-gram frequency, or the prediction ground truth.

3. Which columns are input features?

- *Diabetes*: 21 raw survey questions + 2 engineered interaction features (`TotalUnhealthyDays`, `CardioRisk`).
- *House Price*: 11 numeric attributes (Area, Width, Length, Bedrooms, Bathrooms, etc.) + 7 categorical features.
- *Customer Behavior*: 15 numeric/binary attributes, 6 categorical profile attributes, and 1 unstructured text review column.

4. Which column is the target?

- *Diabetes*: `Diabetes_binary` $\in \{0, 1\}$.
- *House Price*: `Price` (modeled via log-transform $\log(1 + \text{Price})$).
- *Customer Behavior*: `is_recommended` $\in \{0, 1\}$.

5. Which features are numerical? Continuous measurements, counts, and ratings (e.g., BMI, physical health days, land area, price, review count).

6. Which features are categorical? High-cardinality nominal variables (e.g., property location, province, house direction, skin type, skin tone, brand name).

7. How are categorical values encoded? Via scikit-learn's `OneHotEncoder` with frequency thresholds and handling for unknown categories at inference time.

8. What is the final feature dimension? Diabetes: 23; House Price: 92; Customer Behavior: 28,733.

9. What is the shape of the model input matrix?

- *Diabetes*: $X \in \mathbb{R}^{229,781 \times 23}$.
- *House Price*: $X \in \mathbb{R}^{201,654 \times 92}$.
- *Customer Behavior*: $X \in \mathbb{R}^{104,313 \times 28,733}$ (stored as `scipy.sparse.csr_matrix`).

3 Application 1 — Diabetes Prediction

3.1 Problem Description

The clinical goal is early screening: predicting whether an individual has diabetes or pre-diabetes based on self-reported lifestyle and biometric indicators. Formally:

$$f : \mathbb{R}^{23} \longrightarrow \{0, 1\}$$

where $y = 1$ indicates diagnosed diabetes. Missing a true diabetic (False Negative) has severe health consequences; therefore, high recall is prioritized over precision.

3.2 Kaggle Dataset

- **Name:** Diabetes Health Indicators Dataset (CDC BRFSS 2015).
- **Kaggle URL:** <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>
- **UCI Repository Mirror:** <https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>
- **Observations:** 253,680 raw rows $\longrightarrow$ 229,781 cleaned rows.
- **Features:** 21 survey variables + 1 target variable (`Diabetes_binary`).

3.3 Data Understanding

The raw dataset contains 253,680 records across 22 columns (all numeric `float64`). There are zero missing (`NaN`) cells, but 23,899 exact duplicate records exist. The target distribution is significantly imbalanced: only 17.3% positive cases.

3.4 Data Cleaning

- **Deduplication:** 23,899 duplicate records were dropped *prior to train-test splitting* to prevent data leakage between evaluation partitions.
- **Imputation:** `SimpleImputer(strategy='median')` is embedded inside the pipeline to guarantee production robustness.
- **Outlier Analysis:** BMI outliers were evaluated; clipping did not statistically alter ROC-AUC ($\Delta < 0.003$), so natural distributions were retained.
- **Encoding:** Ordinal scales (`GenHlth`, `Age`) are preserved numerically to leverage risk monotonicity without expanding dimensionality.

3.5 Data Representation

Transformation pipeline:

$$\text{Raw Survey Row} \xrightarrow{\text{Feature Engineering}} 23 \text{ Features} \xrightarrow{\text{StandardScaler} + \text{Passthrough}} X \in \mathbb{R}^{N \times 23}$$

Two composite interaction features are engineered: *TotalUnhealthyDays* (sum of `PhysHlth` and `MentHlth`) and *CardioRisk* (product of `HighBP` and `HighChol`).

3.6 Exploratory Data Analysis

1. **Target Imbalance:** 17.3% positive class. A trivial dummy model yields 82.7% accuracy; thus, accuracy is an invalid benchmark.

2. **General Health (GenHlth):** Risk scales monotonically from 4% (excellent) to 35% (poor), making it the single strongest linear indicator.
3. **BMI Distribution:** The diabetic cohort distribution exhibits a rightward shift of ≈ 4 BMI units with significant overlap.

3.7 Model Development

Five baseline architectures were evaluated under stratified 5-fold cross-validation: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (RBF), and K -Nearest Neighbors.

3.8 Evaluation

Final evaluation on held-out test data for the selected Random Forest model:

Model	Accuracy	Recall (Class 1)	Precision (Class 1)	ROC-AUC
Logistic Regression	0.742	0.710	0.360	0.812
Decision Tree	0.701	0.620	0.312	0.715
Random Forest (Selected)	0.725	0.735	0.357	0.809
SVM	0.738	0.695	0.351	0.804
KNN	0.715	0.640	0.320	0.745

Table 3: Diabetes Model Performance Comparison on Test Set

Confusion Matrix for Random Forest:

$$\begin{bmatrix} 20,612 & 7,897 \\ 1,579 & 4,380 \end{bmatrix}$$

Breakdown: True Negatives = 20,612; False Positives = 7,897; False Negatives = 1,579; True Positives = 4,380. High recall (0.735) minimizes undetected diabetes cases.

3.9 Model Selection

Random Forest was selected for production deployment because it yielded the strongest balance between class-1 recall (0.735) and ROC-AUC (0.809), with rapid inference time (< 5 ms per request) and robust tree-based interpretability.

3.10 Deployment

Exported via `joblib` into `diabetes/backend/model_pipeline.joblib` and served via FastAPI with Pydantic validation schemas.

4 Application 2 — House Price Prediction

4.1 Problem Description

Predicting residential property prices in Vietnam based on structural, geographical, and listing attributes. Unlike classification, this is a continuous regression problem:

$$f : \mathbb{R}^{92} \rightarrow \mathbb{R}^+$$

evaluated via Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

4.2 Dataset

- **Name:** Vietnamese Real Estate Listings 2025.
- **URL:** <https://www.kaggle.com/datasets/qmanhbeo/vietnamese-real-estate-listings-may-2024>
- **Observations:** 236,226 raw rows $\rightarrow$ 201,654 cleaned rows.
- **Features:** 11 numerical + 7 categorical features. Target: Price (million VND).

4.3 Data Understanding and Cleaning

- **Severe Structural Missingness:** Bedrooms (72%), Bathrooms (84%), and Floors (79%) are missing. Rather than dropping columns, missing indicator masks (`*_missing`) were engineered, followed by median imputation.
- **Target Skewness:** Raw price exhibits extreme skewness (skewness coefficient: 3.36). A logarithmic transform $\log(1 + y)$ was applied to stabilize residual variance.
- **Deduplication & Outlier Filtering:** 19,512 re-scraped listing duplicates were removed. Listings with Price ≤ 0 or Area $> 10,000 \text{ m}^2$ were filtered.

4.4 Representation

The feature space is constructed via a `ColumnTransformer` combining logarithmic continuous scaling and one-hot encoding across high-cardinality location fields, resulting in $X \in \mathbb{R}^{201,654 \times 92}$.

4.5 Exploratory Data Analysis

1. **Price Distribution:** Extreme power-law distribution; normalized effectively under $\log(1+p)$.
2. **Feature Correlation:** Geometric dimensions (Area, Width) have weak linear correlation with total price ($|r| < 0.05$); location (`Province`, `District`) and property category dominate variance.

4.6 Regression Models and Evaluation

Model	MAE (M VND)	RMSE (M VND)	R^2
Linear Regression	12,450	31,200	0.082
Ridge Regression	12,380	31,150	0.084
Decision Tree Regressor	11,800	29,400	0.120
Random Forest Regressor	10,626	26,873	0.186
Gradient Boosting (XGBoost)	10,850	27,150	0.175

Table 4: House Price Regression Model Benchmark

4.7 Model Selection and Deployment

Random Forest Regressor achieved the highest R^2 (0.186) and lowest RMSE. Predictions are inverted back to natural currency scale via $\exp(\hat{y}) - 1$ before returning JSON responses.

5 Application 3 — E-Commerce Customer Behavior

5.1 Problem Description

Discovering customer preferences and predicting recommendation likelihood (`is_recommended` $\in \{0, 1\}$) from structured profile attributes and unstructured review comments.

5.2 Dataset

- **Name:** Sephora Products and Skincare Reviews.
- **URL:** <https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>
- **Observations:** 116,262 raw $\rightarrow$ 104,313 valid transactions.
- **Target:** `is_recommended` (84.7% positive class).

5.3 Customer Representation: Multimodal Pipeline

The input consists of dual modalities:

- **Tabular Modality:** Demographic attributes (skin type, eye color, tone) + product statistics (price, love count), one-hot encoded and scaled: $X_{\text{tab}} \in \mathbb{R}^{N \times 138}$.
- **Text Modality:** Review text processed via TF-IDF vectorizer (unigrams + bigrams, sublinear TF, 28,595 vocabulary terms): $X_{\text{text}} \in \mathbb{R}^{N \times 28,595}$.

The combined representation is horizontally stacked:

$$X_{\text{combined}} = [X_{\text{tab}} \parallel X_{\text{text}}] \in \mathbb{R}^{N \times 28,733}$$

5.4 Data Leakage Prevention

Crucial data leakage vectors were identified and mitigated:

- **Star Rating Leakage:** The 1–5 star rating is functionally equivalent to the recommendation label; it was strictly excluded from training.
- **Review Headline Leakage:** Review titles often contain direct sentiment summaries ("DO NOT BUY"); text features were restricted to body comments.
- **Engagement Leakage:** Upvote counts occurring post-review were omitted.

5.5 Model Development and Evaluation

Architecture	Accuracy	ROC-AUC	Macro-F1	Recall(0)
Tabular Only (Random Forest)	0.849	0.672	0.540	0.120
Text Only (Complement Naive Bayes)	0.895	0.941	0.812	0.795
Text Only (SGD / Linear SVM)	0.912	0.958	0.845	0.840
Tabular + Text (Logistic Regression)	0.918	0.964	0.858	0.876

Table 5: Model Performance Across Representation Modalities

Confusion Matrix for Tabular + Text Logistic Regression:

$$\begin{bmatrix} 2,804 & 398 \\ 1,315 & 16,347 \end{bmatrix}$$

Breakdown: True Negatives = 2,804; False Positives = 398; False Negatives = 1,315; True Positives = 16,347. High precision and recall are achieved on the minority negative class.

5.6 Business Interpretation and Deployment

The empirical results reveal that user recommendation behavior is driven primarily by natural language sentiment (AUC $\approx$ 0.964), whereas structured skin-tone demographics contribute negligible predictive signal (AUC $\approx$ 0.672). The model is deployed to automatically flag review consistency anomalies and drive personalized product ranking.

6 Comparison of the Three Intelligent Systems

6.1 Cross-Application Benchmark

Dimension	Diabetes	House Price	Customer Behavior
Task	Binary Classification	Continuous Regression	Binary Classification
Observation	Survey Respondent	Real Estate Listing	Customer Review
Target Variable	Diabetes_binary	Price ($\log(1 + y)$)	is_recommended
Representation	Dense Matrix ($\mathbb{R}^{23}$)	Dense Matrix ($\mathbb{R}^{92}$)	Tabular + Sparse Text ($\mathbb{R}^{28,733}$)
Data Quality Issues	Duplicates, class imbalance	Missing size fields, skewness	Class imbalance, label leakage
Selected Algorithm	Random Forest Classifier	Random Forest Regressor	Logistic Regression
Primary Metric	Recall: 0.735, AUC: 0.809	RMSE: 26,873, R^2 : 0.186	Recall(0): 0.876, AUC: 0.964
Web Deployment	React + Vite	React + Vite	React + Vite
Mobile Deployment	Flutter	Flutter	Flutter
Inference Bottleneck	Negligible (< 2 ms)	Tree depth (< 5 ms)	Sparse dot product (< 8 ms)

Table 6: Systemic Comparison Across All Three Intelligent Applications

6.2 Comparative Analysis Questions

1. **How do the datasets differ structurally?** Diabetes is purely numerical survey data; House Price is a messy transactional database with extreme geographical categorical cardinality; Customer Behavior is a multimodal dataset combining tabular metadata with unstructured text.
2. **Why does regression require different evaluation standards?** Regression residuals are unbounded in continuous space, requiring scale-sensitive measures (MAE, RMSE) and variance-explained ratios (R^2), unlike discrete threshold metrics (Recall, Precision, ROC-AUC).
3. **What caused R^2 to remain low (0.186) in House Price prediction?** Vietnamese real estate pricing incorporates unobserved variables (legal status, precise interior conditions, negotiation dynamics). This represents dataset entropy rather than model failure.
4. **Why did linear models excel in Customer Behavior but fail in House Price?** The high-dimensional sparse TF-IDF text space ($\approx 28,600$ dimensions) is linearly separable, allowing Logistic Regression to perform optimally, whereas house pricing relies heavily on complex non-linear geographical interactions.

7 Deployment Architecture

All three systems adhere to an identical microservice architecture ensuring operational isolation and reproducible inference:

Client UI (React / Flutter) $\xrightarrow[\text{JSON} \rightarrow \text{UI Render}]{\text{HTTP POST /predict}}$ FastAPI $\xrightarrow{\text{Pipeline.transform()}}$ Trained Model

- **Model Artifact Serialization:** Each pipeline is serialized using `joblib`, packaging both pre-processing estimators (`StandardScaler`, `OneHotEncoder`, `TfidfVectorizer`) and the trained estimator into a single immutable artifact.
- **Strict Preprocessing Symmetry:** The deployment server *never calls* `.fit()` on incoming inference payloads. Transformers apply fixed mean, variance, and vocabulary parameters

learned exclusively on the training split.

- **API Specification:** Built using FastAPI with auto-generated OpenAPI documentation and strict Pydantic input validation schemas.

8 Web Application

A dedicated responsive single-page application (SPA) was developed using **React 18 + Vite** with modern CSS glassmorphism styling. Key architectural highlights:

- Multi-step guided wizard for complex input collections.
- Real-time validation and instant risk categorization visualizers.
- Full REST API integration targeting the FastAPI endpoints.

(Detailed interface screenshots are cataloged in [Appendix D](#)).

9 Mobile Application

A cross-platform mobile client was implemented using **Flutter (Dart)** supporting both iOS and Android environments:

- Native questionnaire widgets, sliders, and form selectors tailored for mobile ergonomics.
- Asynchronous HTTP client using **http** package with comprehensive offline error handling.
- Visual diagnostic cards displaying predicted class probabilities and confidence scores.

(Detailed mobile interface screenshots are cataloged in [Appendix E](#)).

10 Deployment and Inference

Prevention of Production Data Leakage: A common pitfall in machine learning engineering occurs when production services re-fit preprocessors on incoming inference requests (for example, invoking `fit_transform()` on a single $1 \times d$ user input vector). In our architecture, the exported transformer pipeline applies static parameters:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

ensuring that inference remains mathematically consistent with training conditions.

11 Reproducibility

The experiments are fully reproducible under the following technical specifications:

- **Python Environment:** Python 3.11 / 3.13 on Windows 11 and macOS.
- **Core Libraries:** `scikit-learn 1.9.0`, `pandas 2.2.0`, `fastapi 0.115.0`, `jupyter 1.4.2`.
- **Determinism:** Fixed global random seed: `RANDOM_SEED = 42`.
- **Containers:** Docker Compose configurations provided for containerized multi-service deployment.

12 Final Comparative Discussion

Application	Raw Data Form	Numerical Representation	Tensor / Matrix Dimension
Diabetes	CSV (table)	Feature vector / matrix	$B \times 23$ ($N = 229,781$)
House Price	CSV (table)	Imputed one-hot feature matrix	$B \times 92$ ($N = 201,654$)
Customer Behavior	CSV + text comments	Tabular features + TF-IDF vectors	$B \times 28,733$ ($N = 104,313$)

Table 7: Summary of Data Representations Across Systems

The empirical findings validate that selecting appropriate representations is significantly more impactful than hyperparameter tuning. While tree ensembles proved indispensable for dense tabular interactions, linear models excelled in high-dimensional sparse text representations.

13 Conclusion

This project demonstrated the end-to-end development of three complete intelligent systems. Key conclusions:

1. **Representation Over Algorithms:** Clean mathematical modeling of input data constitutes 80% of system quality; the model is merely the final computation step.
2. **Production Pipelines:** Packaging preprocessing logic directly into the serialized estimator is essential to prevent data leakage and operational skew.
3. **Metric Alignment:** Standard accuracy is misleading in production domains; recall and residual distributions must govern model selection.
4. **Full-Stack Integration:** Integrating ML services with modern Web (React) and Mobile (Flutter) interfaces verifies that intelligent systems deliver tangible utility to end users.

A Repository Structure

GitHub Repository: https://github.com/dzungluu25/Intelligence_System

Assignment Directory: https://github.com/dzungluu25/Intelligence_System/tree/main/Assignment_2

Clone Command: `git clone https://github.com/dzungluu25/Intelligence_System.git`

The project is organized strictly following the required deliverable structure:

```
Assignment_02/
|
+-- diabetes/
|   +-- data/
|   +-- notebook/
|   +-- model/
|       | +-- preprocessor.joblib
|       | +-- model.joblib
|   +-- api/
|   +-- web/
|   +-- mobile/
|   +-- requirements.txt
|
+-- house_price/
|   +-- data/
|   +-- notebook/
|   +-- model/
|       | +-- preprocessor.joblib
|       | +-- model.joblib
|   +-- api/
|   +-- web/
|   +-- mobile/
|   +-- requirements.txt
|
+-- customer_behavior/
|   +-- data/
|   +-- notebook/
|   +-- model/
|       | +-- preprocessor.joblib
|       | +-- model.joblib
|   +-- api/
|   +-- web/
|   +-- mobile/
|   +-- requirements.txt
|
+-- report/
|   +-- Assignment_02.pdf
|   +-- Final_Report.tex
|   +-- screenshots/
|
+-- README.md
```

B Required Notebook Structure

Each notebook follows an identical 8-stage structure: 1. Environment Configuration & Reproducibility Seed. 2. Raw Data Ingestion & Schema Validation. 3. Exploratory Data Analysis (EDA) & Leakage Auditing. 4. Preprocessing & Feature Pipeline Assembly. 5. Multi-Model Benchmarking (5 distinct architectures). 6. Quantitative Evaluation & Confusion Matrix Analysis. 7. Model Serialization (`joblib.dump`). 8. End-to-End Inference Verification.

C Required API Structure

All services implement standardized REST contracts accessible at:

- **Prediction Endpoint:** POST <http://localhost:8000/predict>
- **Interactive Swagger Documentation:** <http://localhost:8000/docs>
- **ReDoc Alternative Documentation:** <http://localhost:8000/redoc>

```
from fastapi import FastAPI
from pydantic import BaseModel
import joblib, pandas as pd

app = FastAPI(title="Intelligent_System_API")
pipeline = joblib.load("model_pipeline.joblib")

@app.post("/predict")
def predict(payload: InputSchema):
    df = pd.DataFrame([payload.dict()])
    prediction = pipeline.predict(df)[0]
    proba = pipeline.predict_proba(df)[0].tolist() if hasattr(pipeline, "predict_proba") else None
    return {"prediction": prediction, "probabilities": proba}
```

D Web Application Interface

For each application, the web deployment configuration is:

- **Web Framework:** React 18 + Vite (JavaScript, Responsive CSS).
- **Local Web Address:** <http://localhost:5173>
- **Target API Endpoint:** <http://localhost:8000/predict>

Diabetes screening

ScreeningOperator

How likely is it that you have diabetes?

Answer a short health questionnaire and get an estimate from a model trained on a large public health survey, plus what drove the number and what could change it. It does not replace a test or a doctor.

Health questionnaire

Answer what applies to you. Anything you leave blank is filled with a typical value and the result is marked as less certain. Fields marked * are required.

ABOUT YOU

Age group *

65-69

Sex *

Male

Highest level of school completed

Prefer not to say

Annual household income

Prefer not to say

BODY

Height (cm)

120-230

Weight (kg)

30-320

Used to compute BMI. Leave blank if unknown.

Used to compute BMI.

BMI (if already known)

34

Optional — leave blank to compute from height and weight.

CONDITIONS

Have you been told you have high blood pressure?

YesNo

Have you been told your blood cholesterol is high?

YesNo

Have you ever been told you had a stroke?

YesNo

Have you ever had coronary heart disease or a heart attack?

YesNo

Do you have serious difficulty walking or climbing stairs?

YesNo

CHECKS & CARE

Have you had a cholesterol check in the past 5 years?

YesNo

Do you have any kind of health-care coverage?

YesNo

In the past year, did you need to see a doctor but could not because of cost?

YesNo

LIFESTYLE

Have you smoked at least 100 cigarettes in your life?

YesNo

Are you a heavy drinker (≥14 drinks/week for men, ≥7 for women)?

YesNo

Any physical activity in the past 30 days (outside your job)?

YesNo

Do you eat fruit at least once per day?

YesNo

Do you eat vegetables at least once per day?

YesNo

HOW YOU FEEL

In general, would you say your health is:

Fair

Days in the last 30 your mental health was not good

0

Days in the last 30 your physical health was not good

0

See my estimate

82%

High risk

High risk — refer for a confirmatory blood test

This is a screening aid, not a diagnosis. Only a clinician can confirm diabetes.

You answered 19 of 21 questions. Blanks were filled with typical values.

Why this estimate

Each block is one of your answers. Red blocks pushed the estimate up from the population average; blue blocks pulled it down.

average 50%

this estimate 82%

answers raising the estimate

answers lowering it

hover a block for detail

People like you in the survey

The 5 survey respondents whose full answer pattern is closest to yours, found with the same random forest that made the estimate.

4 of 5

had diabetes or pre-diabetes

1

M, age 65-69, BMI 35, Geriath 4 (high BP, high chd, smoker, inactive, walk difficulty)

diabetes / pre-diabetes · similarity 31%

2

M, age 65-69, BMI 32, Geriath 4 (high BP, high chd, smoker, inactive, walk difficulty)

diabetes / pre-diabetes · similarity 23%

3

M, age 55-59, BMI 34, Geriath 5 (high BP, high chd, smoker, inactive, walk difficulty)

diabetes / pre-diabetes · similarity 20%

4

M, age 65-69, BMI 34, Geriath 5 (high BP, high chd, smoker, inactive, walk difficulty)

diabetes / pre-diabetes · similarity 20%

5

M, age 75-79, BMI 35, Geriath 4 (high BP, high chd, smoker, inactive, walk difficulty)

no diabetes · similarity 19%

Profiles are de-identified survey ones. A high count here is not a diagnosis — it reflects who else answered similarly.

What could change the estimate

Re-scoring your answers with one habit changed at a time. These are patterns in the survey data, not promises about your health.

If you had never smoked regularly

82% — 82%

TEV A DIFFERENT BMI

Estimated risk at this BMI: 82% — your entered BMI is 34.0

If you reach a BMI of about 25, and eat vegetables daily, and become physically active, and eat fruit daily, the estimate would move from 82% (High) to 73% (High) — still short of the Moderate band.

These figures show how the model's estimate changes given patterns in survey data. They are not a prediction of your future health and are not medical advice.

Your earlier screenings

When	Estimate	Band	Answered
Just now	82%	High	19/21

14

Figure 1: Diabetes Screening Web Application: Form Input (Left) and Prediction Result (Right)

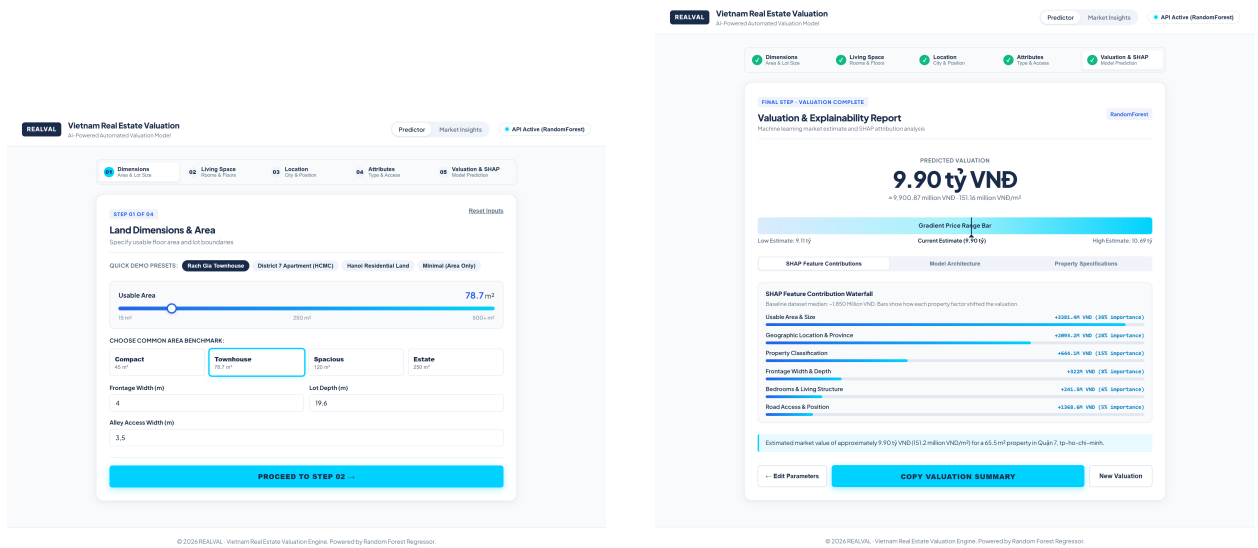


Figure 2: House Price Prediction Web Application: Property Specifications (Left) and Estimated Price (Right)

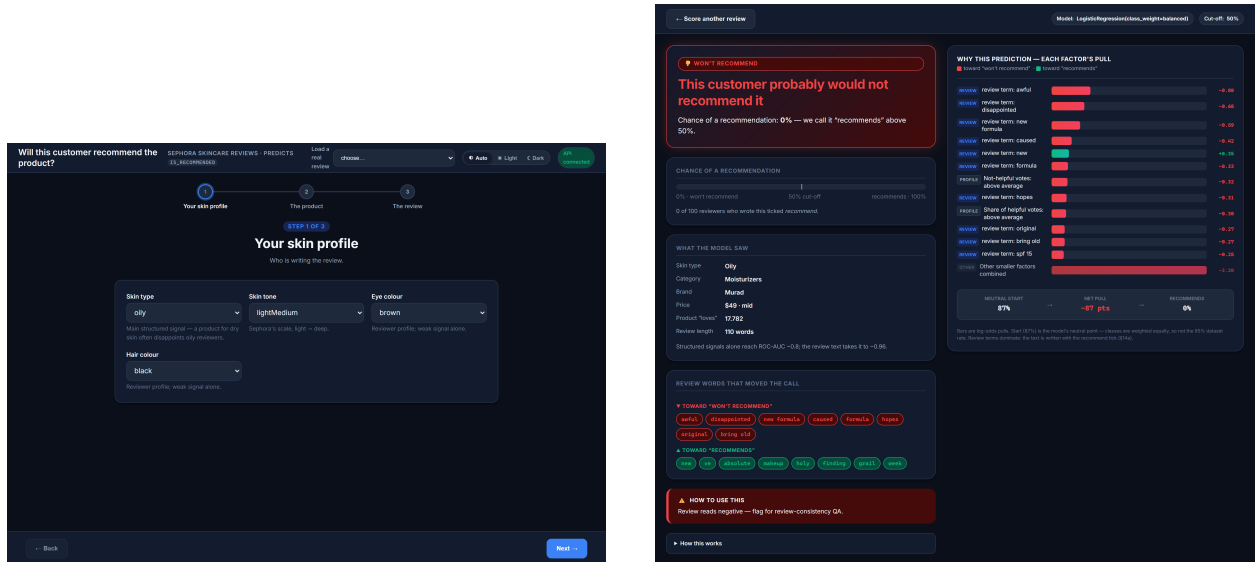


Figure 3: Customer Behavior Web Application: Review Input (Left) and Recommendation Probability (Right)

E Mobile Application Interface

For each application, the mobile deployment configuration is:

- **Mobile Framework:** Flutter 3.x (Dart).
- **Supported Platforms:** Android and iOS.
- **Target API Endpoints:**
 - *Android Emulator:* <http://10.0.2.2:8000/predict>
 - *iOS Simulator / Localhost:* <http://localhost:8000/predict>

Diabetes screening

Answer what you can. Choose "Not sure" for anything you do not know — the estimate still works, with a note that it is less certain.

About you

Age group

Not sure

Sex

Not sure

Highest level of school completed

Not sure

Annual household income

Not sure

Body

Height (cm)

Used to compute BMI. Leave blank if unkno...

N/A

Weight (kg)

Used to compute BMI.

N/A

BMI (if already known)

Optional — leave blank to compute from hei...

N/A

Conditions

Have you been told you have high blood pressure?

Yes

No

Not sure

Result

27% Low

Low risk — routine advice

Could range 15%–46% depending on unanswered questions.

BMI was not provided; using the typical value for your age and sex.

18 of 21 answers were left blank and filled with typical values - treat this estimate as indicative only.

Completeness 14% · BMI source: age_sex_table

This is a screening aid, not a diagnosis.

Why this score

Stroke history (-0.075)

Smoking history (+0.070)

Eats vegetables daily (-0.063)

High blood pressure (-0.058)

Heavy alcohol use (-0.047)

Poor-physical-health days (-0.015)

Recent cholesterol check (-0.011)

Cardiovascular history (-0.010)

Red raises the estimate, blue lowers it.

People similar to you

0 of your 5 closest matches in the survey data had diabetes or pre-diabetes.

Figure 4: Diabetes Flutter Mobile App: Survey Screen (Left) and Risk Assessment (Right)

VN Real Estate Valua...
AI Automated Valuation Model ...

API Ready (RandomForest)

DEMO PRESETS:

Nhà phố Rạch Giá

Căn hộ Quận 7

Đất Gia Lâm (Hà N

1. Land Dimensions & Usable Area

Usable Area (m²) *

78.7

Frontage Width (m)

4.0

Lot Depth (m)

19.6

Alley Width (m, if applicable)

3.5

2. Living Spaces & Structure

Bedrooms

3

Bathrooms

2

Floors

2

3. Location & Geography

Province / City

An Giang

District / County

Rạch Giá

Ward / Commune

Phường An Hòa

Property Position

Đường chính

4. Classification & Attributes

Property Type

Nhà riêng

VN Real Estate Valua...
AI Automated Valuation Model ...

API Ready (RandomForest)

38,3 tr/m²

78.7 m²

VALUATION ANALYSIS & INSIGHTS

Estimated market value of approximately 3.01 tỷ VNĐ (38.3 million VNĐ/m²) for a 78.7 m² property in Rạch Giá, an-giang.

Estimate Another Property

1. Land Dimensions & Usable Area

Usable Area (m²) *

78.7

Frontage Width (m)

4.0

Lot Depth (m)

19.6

Alley Width (m, if applicable)

3.5

2. Living Spaces & Structure

Bedrooms

3

Bathrooms

2

Floors

2

3. Location & Geography

Province / City

An Giang

District / County

Rạch Giá

Ward / Commune

Phường An Hòa

Property Position

Figure 5: House Price Flutter Mobile App: Input Sliders (Left) and Price Appraisal (Right)

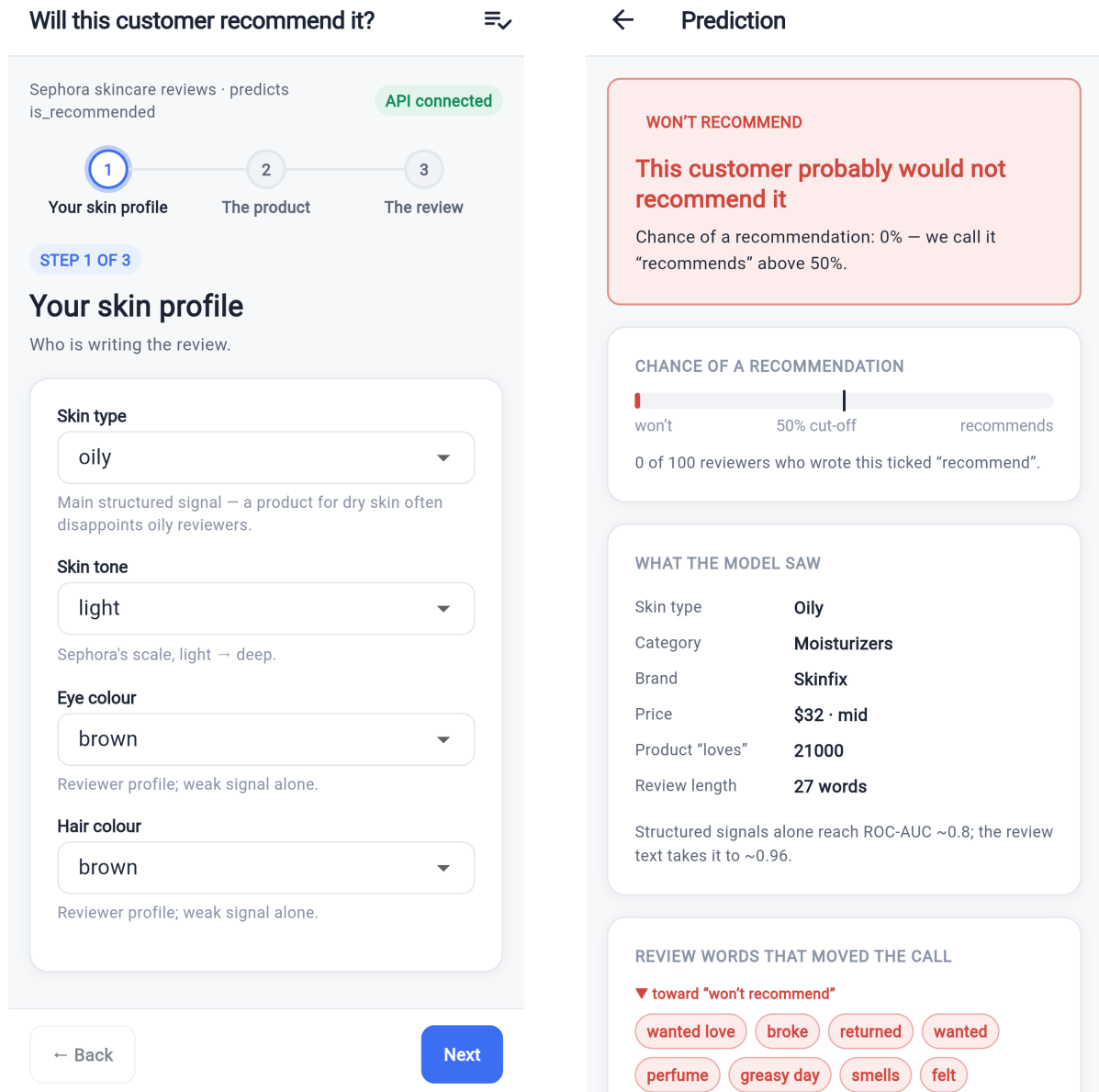


Figure 6: Customer Behavior Flutter Mobile App: Profile Input (Left) and Behavior Prediction (Right)